SQLAlchemy

Requirements:

Flask
flask_sqlalchemy
SQLAlchemy

Imports:

- `from flask import Flask`
- `from flask_sqlalchemy import SQLAlchemy`
- `from sqlalchemy.orm import DeclarativeBase, Mapped, mapped_column`
- `from sqlalchemy import Integer, String, Float`

Configuring:

```
# create the app
app = Flask(__name__)

# configure the SQLite database, relative to the app instance folder
app.config["SQLALCHEMY_DATABASE_URI"] = “sqlite:///project.db"

# initialize the app with the extension
db.init_app(app)
```

Creating table:

```
#creating the table
class Book(db.Model):
    id: Mapped[int] = mapped_column(Integer, primary_key=True)
    title: Mapped[str] = mapped_column(String(250), unique=True, nullable=False)
    author: Mapped[str] = mapped_column(String(250), nullable=False)
    rating: Mapped[float] = mapped_column(Float, nullable=False)

    # Optional: this will allow each book object to be identified by its title when printed.
    def __repr__(self):
        return f'<Book {self.title}>'

# Create table schema in the database. Requires application context.
with app.app_context():
    db.create_all()
```

CRUD Operations

CREATE Records

```
# CREATE RECORD
with app.app_context():
    new_book = Book(id=1, title="Harry Potter", author="J. K. Rowling", rating=9.3)
    db.session.add(new_book)
    db.session.commit()
```

READ Records

- `with app.app_context():`
- `result = db.session.execute(db.select(Book).order_by(Book.title))`
- `all_books = result.scalars()`

To read all the records we first need to create a "query" to select things from the database. When we execute a query during a database session we get back the rows in the database (a `Result` object). We then use `scalars()` to get the individual elements rather than entire rows.

READ particular Record

- `with app.app_context():`
- `book = db.session.execute(db.select(Book).where(Book.title == "Harry Potter")).scalar()`

To get a single element we can use `scalar()` instead of `scalars()`.

Update A Particular Record By Query

- `with app.app_context():`
- `book_to_update = db.session.execute(db.select(Book).where(Book.title == "Harry Potter")).scalar()`
- `book_to_update.title = "Harry Potter and the Chamber of Secrets"`
- `db.session.commit()`

Update A Record By PRIMARY KEY

- `book_id = 1`
- `with app.app_context():`
- `book_to_update = db.session.execute(db.select(Book).where(Book.id == book_id)).scalar()`
- `# or book_to_update = db.get_or_404(Book, book_id)`
- `book_to_update.title = "Harry Potter and the Goblet of Fire"`
- `db.session.commit()`

Delete A Particular Record By PRIMARY KEY

- `book_id = 1`
- `with app.app_context():`
- `book_to_delete = db.session.execute(db.select(Book).where(Book.id == book_id)).scalar()`
- `# or book_to_delete = db.get_or_404(Book, book_id)`
- `db.session.delete(book_to_delete)`
- `db.session.commit()`